SYSTEM ANALYSIS DOCUMENTATION (SAD)

UMHackathon 2026

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1. Introduction:

This section introduces KaOne, the strategic AI-driven solution designed to address the financial management challenges for Malaysian SMEs identified in the Product Requirement Documentation.

1.1. Purpose:

The system analysis document highlights the technical scope and design decisions behind the development of KaOne. The key elements of the system covered are as follows:

So, the key element of system that has been covered are as followed:

1. **Architecture**:
    a. KaOne is built as an AI-Native platform integrated with the Gemini Pro engine for specialized financial reasoning
    b. The frontend is designed as a conversational workspace for managing financial inputs via text, images, or voice.
2. **Data Flows:**
    a. Unstructured data, such as CSV, receipts, or prompts, flows from the user interface to the Gemini Pro model.
    b. The model parses the intent, identifies financial relationships, and returns structured insights such as tax predictions or budget updates to the dashboard.
3. **Model Process:**
    a. The system utilizes multi-step agentic prompting to handle end-to-end workflows: from extracting transaction data to validating mathematical accuracy and generating strategic recommendations
4. Role of Reference:
    a. This document serves as a technical reference for developers and testers to ensure consistent implementation of the AI Financial Intelligence features and scope.

1.2. **Background**:

Small business owners in Malaysia, such as home-based bakers, often rely on manual bookkeeping or rigid software that does not account for local tax nuances (LHDN). This manual process creates a barrier for entrepreneurs trying to visualize their business/

1. Previous Version:
    a. The initial concept was focused on general UI/UX design scaffolding through natural language.
    b. Existing tools required users to manually split personal and business spending, a process prone to labor-intensive errors.

2. Changes in Major Architectural Components
    a. Shift from a general design engine to a specialized financial intelligence engine utilizing the Gemini Pro model for local tax and expense accuracy.
    b. Integration of OCR (Optical Character Recognition) capabilities to support Picture-to-Finance inputs alongside standard text prompts.
3. New capabilities that are only introduced in this
    a. Gemini Pro-Powered Forecasting: Real-time prediction of revenue and tax savings based on historical data.
    b. Subscription Kill Switch: An automated agent that flags unused services and drafts cancellation scripts.
    c. Smart transaction splitter: A simplified interface for classifying mixed-use spending with a single click

1.3. Target Stakeholder

Stakeholders	Roles	Expectations
Small Business Owner	Interacts with the AI Analyst via natural language to oversee cash flow, manage inventory and predict tax obligations.	A simple, non-technical financial experience with automated "Pocket CFO" insights and bilingual support (English/Bahasa Melayu).
AI Analyst (Gemini Pro)	Acts as a reasoning engine to parse unstructured data, classify transactions, and provide predictive forecasting.	High-fidelity data grounding and accurate financial parsing based on localized Malaysian tax rules.
Development Team	Responsible for building and maintaining the modular architecture, integrating the Gemini Pro API, and ensuring Firestore data persistence.	Clear API usage contracts for the ILMU engine and a well-documented, modular codebase using Vanilla JS and CSS Variables
QA Team	Validates system behavior across all features, including the "Kill Switch" negotiation and Smart Tax Forecaster.	Defined scope, robust validation of API request/response formats, and coverage of edge cases like API failure (Graceful Degradation).
LHDN (Indirect Stakeholder)	The regulatory body for which the system prepares	Accurate estimation of tax targets based on current Malaysian tax rates and

	tax estimation data for the business owner.	proper business expense classification.

2. System Architecture & Design

2.1. High Level Architecture

2.1.1. Overview:

Type	Details
System	Web-based
Architecture	AI-Native Client-Server Architecture

kaOne is structured as an AI-integrated web application designed for a single-user business owner dashboard. The system enables the user to upload financial data which is then processed through an API layer to the core intelligence engine. The architecture consists of three primary services: the Financial Analysis Service, the Inventory Management Service, and the Notification/Alert Service.

2.1.2 LLM as Service Layer

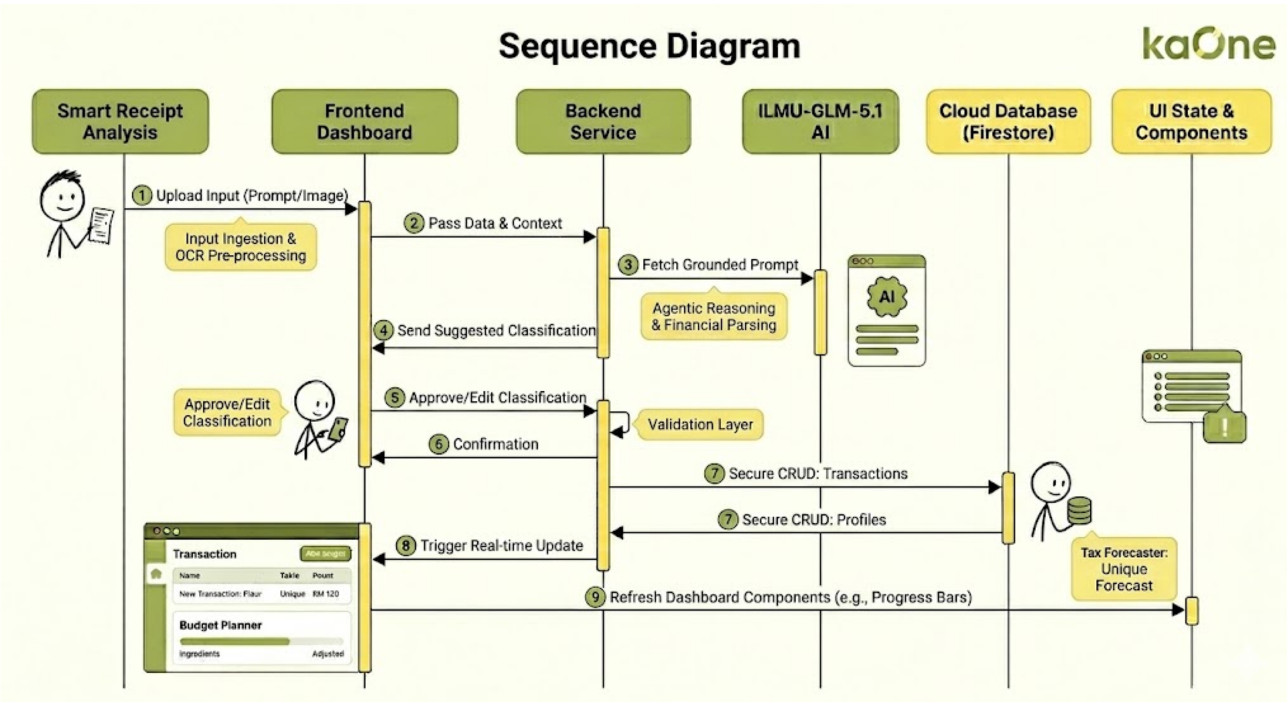
The **Gemini Pro** is integrated as a core service layer rather than a standalone module. It acts as the brain of the system, handling reasoning tasks such as tax prediction and expense optimization:

2.1.3 Dependency Diagram

a. Prompts are constructed using multi-step agentic prompting. The system first identifies intent, then grounds the prompt with specific user data (e.g., current bank balance or inventory levels) before sending it to the Gemini Pro API.
b. The context window includes the current business profile, a rolling history of recent transactions and current inventory stock.
c. The backend receives raw responses such as JSON or text from the IMU API. A dedicated Validation Agent parses the response for financial tokens (RM values, percentages), validates the math and passes the structured data to the frontend to update specific dashboard components.
d. Token limitations are enforced at the Backend API layer. If an input like a large CSV exceeds the 32k token limit, the system performs intelligent chunking batching the data chronologically before it is sent to the Gemini Pro.
e. Major API key & System Interactions:
    - The Web Dashboard calls Firebase Authentication to manage secure user sessions and login states.

- The system uses the Firebase SDK to perform CRUD operations (Create, Read, Update, Delete) on transactions, inventory, and chatHistory collections to ensure data persistence.
- The system initiates secure HTTPS POST calls to the https://api.ilmu.ai/v1/chat/completions endpoint for all AI reasoning tasks

2.2.2. Sequence Diagram



2.2. Technological Stack

2.2.1. Frontend

- Technologies: HTML5, CSS3 and JavaScript
- Chosen for high-speed rendering of the functional mockup and conversational dashboard. The use of CSS Custom Properties (Variables) ensures a coherent and consistent design flow across the entire system

2.2.2. Backend

- Technologies: JavaScript/Node.js environment integrating the Firebase SDK and Fetch API.
- Manages the service layer orchestrating calls between the frontend dashboard and the Gemini Pro intelligence engine for financial reasoning.

2.2.3. Database

- Technologies: Google Firebase Firestore
- A cloud-native NoSQL document database used to ensure data persistence for user profiles, transaction history, inventory and chat logs.

2.2.4. Cloud/Deployment

- Technologies: Firebase Hosting and Google firebase Cloud Infrastucture
- Provides secure user authentication through Firebase Auth and managed cloud storage for real-time synchronization across devices.

2.3. Key Data Flows

2.3.1. The system ensures data integrity by adhering to proper data flow laws avoiding Black Holes where data enters but never leaves.

2.3.1.1. Data Flow Diagram (DFD)

a. Process 1.0: Ingestion & OCR
   - Source: The Small Business Owner (External Entity) uploads a receipt or types a prompt.
   - Transformation: The system captures the raw input. If it's a photo, the OCR engine extracts the text.
   - Output: "Raw Financial Text" is sent to the next process.

b. Process 2.0: AI Reasoning (Gemini Pro)
   - Input: Raw text and the User's Profile (context).
   - Transformation: The AI identifies the intent and parses the text into "Financial Tokens" structured data like amount: 120.00, category: Ingredients and date: 2026-04-26.
   - Law Check: No "Black Hole" exists; every input results in a categorized output.

c. Process 3.0: Persistence (Firestore)
   - Input: Structured Financial Tokens.
   - Transformation: The system writes this data into the Transactions and Inventory collections in Firestore.
   - Data Store: This ensures the information is "persisted" which is saved for later retrieval and doesn't just vanish after the chat session.

d. Process 4.0: Visualization
   - Source: Firestore Data Store.
   - Transformation: The frontend "fetches" the saved records.
   - Final Destination: Data is rendered as Stat Cards, Progress Bars, and Table Rows on the User's Dashboard.

2.3.2. **Normalized Database Schema**

kaOne utilizes a relational data model within Firestore to maintain 3rd Normal Form (3NF) principles for data integrity:

a. **User Collection:** Primary entity storing unique userId, business profile (e.g., Home Bakery), and settings.

b. **Transactions Collection:** Each document is an atomic record linked to a userId containing date, description, amount, and categoryId.

c. **Inventory Collection:** Relational records tracking physical stock (e.g., Flour, Sugar) with a 1:N relationship to the user.
d. **Categories Table:** Acts as a lookup reference for the **Budget Planner**, resolving relationships between raw transactions and higher-level budget buckets like "Ingredients" or "Marketing".

3. Functional Requirements & Scope

This section highlights the core boundaries of the projects. That enables them to focus on a high-impact MVP to showcase the technical feasibility of the project.

3.1. Minimum Viable Product:

3.1.1. This section lists down core features that KaOne will build and demonstrate by the end of UMHackathon 2026.

#	Features	Description
1	Dashboard and Transaction Overview	A central home screen displaying the user's current balance, monthly income, monthly expenses, and Gemini Pro's estimated tax to save. It also includes a Recent Transactions table with business or personal categorization and a Gemini Pro Financial Insight Banner that updates recommendations to users based on the prior three months of data.
2	AI Analyst Chatbot	A conversational AI Interface powered by Gemini Pro, where users can ask questions related to finance or select from the Quick Questions Panel. It also analyzed and read by uploaded documents (CSV, PDF, Excel), receipts (PNG or JPEG) that will go through OCR Processing, as well as receiving personalized financial advice based on the actual data.
3	Inventory and Stock Management	A real-time inventory tracker tailored for micro-business owners like home bakers, housewives, and bachelors to log ingredients and supplies with quantities and values. In addition, it also flags low-stock items and supports manual add and update actions.
4	Budget Planner	A visual budget monitoring tool that displays spending progress bars across categories. For example, Ingredients, Marketing, Delivery, Utilities, and Subscriptions. The progress bar is set against

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		the user’s set limits, with a color code that indicates the expenses' state.
5	Gemini Pro Financial State Forecast	An AI-powered forecasting card that uses Gemini Pro to predict next month’s revenue, expenses, and recommended tax savings based on the user’s last three months of data. The user can click Refresh to get a new prediction after new data is entered.

3.2. Non-Functional Requirements (NFRs)

This section highlights the system qualities that are not features but are crucial for reliability and improving the system.

Quality	Requirements	Implementation
Performance	The dashboard and transaction data should load quickly upon login without noticeable delay	Page load time under 3 seconds on standard broadband connection.
AI Response Time	The Gemini Pro AI Analyst should return a response to user queries within an acceptable timeframe.	Chat response generated within an estimated 5 seconds for text queries and document analysis within an estimated 15 seconds
Accuracy	The Gemini Pro Forecast predictions for revenue, expenses, and tax savings should be based on real user data and produce reliable estimates	Forecast deviation within a range of 15% of the actual monthly figures based on three months of data
Security	User financial data, transaction history, and personal information must be protected from unauthorized access.	Password hashed, API Keys are hidden, no sensitive data exposed in client-side mode.
Usability	The platform should be convenient enough for non-technical users, such as home bakers or micro-business owners with no accounting knowledge.	New users can complete core tasks such as adding new inventory, reading forecasts, and chatting with AI without needing a tutorial or an onboarding guide.
Availability	The system should remain accessible throughout the hackathon demo period without downtime.	99% uptime during UMHackathon 2026 demonstration window.
Scalability	The system uses Firebase as the cloud backend, supporting growth beyond the hackathon demo without requiring a full rebuild.	Firebase handles authentication, real-time database sync, and storage.
Receipt Parsing Accuracy	Uploaded receipt photos and documents should be correctly interpreted by the AI with minimal errors.	Correct extraction of item name, quantity, and cost from at least 80% of uploaded receipts.

3.3. Out of Scope / Future Enhancements

This section shows features that are part of KaOne's grand vision but will not be built or demonstrated during UMHackathon 2026. They are planned for future development during the final or post-hackathon.

a. Voice Input for AI Analyst – The system allows users to speak naturally to log expenses, ask financial questions, or update inventory hands-free (currently visible in the UI as "coming soon")

b. Full LHDN Tax Filing Integration – The system automatically generates and submits tax filing documents directly to Malaysia's LHDN portal based on the user's tracked income and expenses.

c. Mobile Application (iOS & Android) – A native mobile version of KaOne, allowing users to manage finances, scan receipts, and chat with the AI Analyst easily.

d. Multi-business Profile Support - The system allows a single user to manage multiple business accounts. For example, a user runs both a home bakery and a freelance service under one account.

e. Automated Bank Statement Sync – The system will integrate directly to the Malaysian banks like Maybank, CIMB, and RHB to automatically import and categorize transactions without manual entry.

f. Supplier and Invoice Management – A dedicated module for managing supplier contacts, sending invoices to customers, and tracking payment statuses.

g. Team and Collaborator Access – The system allows business owners to invite a business partner or accountant to view or co-manage the financial dashboard with role-based permissions.

h. Spending Limit Alerts & Notifications – The system will automatically push notifications via email or SMS when a budget category is approaching or has exceeded its limit to the users.

i. Historical Reports and Export – The system will generate monthly or yearly financial reports in PDF or Excel format that can be downloaded by users for record keeping or sharing with accountants.

j. Loyalty and Rewards Tracking – For business owners that sell directly to customers, the system can track repeat buyers through the same bank account transaction and manage a simple loyalty point system within the platform.

4. Monitor, Evaluation, Assumptions & Dependencies

4.1. Technical Evaluation:

4.1.1. Grayscale Rollout & A/B Testing:

Given that kaOne is developed in a limited time period of UMHackathon 2026, our team, Winx, conducted the grayscale rollout and A/B testing internally between the group members rather than external users that we don't have yet.

Grayscale Rollout:

As each feature of kaOne is developed, it is immediately tested by our team in a staged manner rather than waiting for all features to be completed. For example, when the Gemini Pro Forecast feature has finished coding, it is first tested by the team member who developed it, and then later it is handed to the remaining members to test on their own devices and environments. This fosters a small-scale rollout where different "users" interact with the feature under different conditions, like different browsers, screen sizes, as a laptop can be 15 inches or 16 inches, and data inputs.

If any issues can be detected at any stage, we will take rapid action to fix them and retest before the features are considered good and stable enough to integrate into the main build. The features that pass during the internal staged view will be demonstrated at this preliminary round.

A/B Testing:

For certain UI and UX decisions, our team conducts a simple internal A/B comparison by preparing two types of features' looks during development. For example, two layouts of the Dashboard summary cards are built and are evaluated by all members. It is to determine which shows financial information clearly and is easier to read by users. Afterwards, we collected all the feedback from each of us, and then the preferred variant was adopted as the final design for the dashboard.

While this is not a large-scale A/B test, it still serves the same purpose within the preliminary round context that will ensure the design and feature decisions are based on observed preferences rather than just making assumptions.

4.1.2. **Emergency Rollback (ER) & Golden Release**:

Within the timeframe of preliminary rounds, to prevent system crashes during live judging or final demonstrations, our team executed Emergency Rollback (ER) and Golden Release.

Emergency Rollback (ER):

Emergency Rollback plays a very crucial role during the development process. For example, when a newly added feature like Firebase authentication fails, or the data fails to be read by the AI, it all happens, which introduces a critical issue. The Emergency Rollback (ER) is triggered. The one in charge of the code will immediately revert the code to the last known Golden release using version control, then the application will be restored to its last stable state.
This is very important for us during the development process, as the time pressures exerted on us may lead to rushed changes. Instead of attempting to fix the errors under pressure, we all agree to roll back to the previous stable state and reintroduce the problem feature if time permits and the fix is confirmed safe.

Golden Release:
For each of the features that are tested and showing positive feedback during our internal staged testing, that version is committed and considered the Golden Release in our progress. This becomes the safe fallback point that the team can return to at any time. The Golden Release is updated when only the newer version has been tested and confirmed stable by all members. This is to ensure that we operate in a reliable working building available, especially in the critical hours before the demonstration video is recorded.

4.1.3. **Priority Matrix:**

The following matrix defines the severity levels, trigger conditions, and corresponding actions that our team, Winx, takes during the development process.

Priority Level	Severity	Trigger Conditions	Action Taken
P1-Critical	The system is completely unusable	Firebase authentication fails, and users cannot log in, or even the dashboard fails to load when the website is run.	Immediately trigger Emergency Rollback to the last Golden Release and halt all new features until resolved.
P2-High	The core feature is broken	Gemini Pro Forecast returns incorrect or null values, or the receipt and document uploaded by the users fail to be parsed by the AI.	Assigned to the developer responsible for that feature immediately, and if it is unresolved within the time limits we set, like 30 minutes, then rollback that specific feature to its last stable state.
P3-Medium	The features are degraded but still usable	Budget Planner progress bars display incorrect values, or the AI Analyst response took longer than expected	Logged and prioritized for fixing before the demo, and a temporary workaround was applied if available.
P4- Low	Minor visual or non-critical issue	UI misalignment on certain screen sizes, incorrect color coding on the inventory low-stock indicator, or minor typos in AI responses	Logged and addressed only if we have more time left after all critical issues are addressed and resolved.
P5- Negligible	Cosmetic issue with no functional impact	Minor spacing inconsistencies, font sizes misaligned, or non-critical notification wording	Acknowledge, but will not be actioned during the limited time, but will be resolved whenever got free time

4.2. **Monitoring**:

This section shows the planning for our monitoring plan of the system and how it will trigger if any damage is detected in the health of the system of kaOne during the preliminary round.

#	Area Monitored	Health Indicators	Unhealthy Trigger	Action Taken
1	User Authentication	Team members can create an account and successfully log in onto all tested devices.	Repeated login failures across devices.	Check the Firebase Authentication console as well as the coding for it. Then fix the configuration and escalate to P1 if unresolved.
2	Database Read and Write	Data entered by users is saved and retrieved correctly in Firebase's storage.	Data not appearing after submission, or incorrect values are displayed	Review Firebase security rules and database structure, and revert the last change if needed.
3	Receipt Upload	Uploaded receipt photos are processed, and data is extracted correctly through OCR processing.	Upload fails, or AI returns empty extraction results.	Check the Firebase storage connection and re-test with a cleaner image and better resolution, or disable the feature for a while if P2 is triggered.
4	Gemini Pro Forecast	The forecast card displays the correct	Null values or obviously incorrect and	The developer responsible for the forecast

		predicted revenue, expenses, and tax savings without errors.	ridiculous predictions were returned.	module will investigate immediately where the root cause is.
5	AI Analyst Chat	Responses are relevant and returned within an acceptable time.	Response takes too long or returns irrelevant or false content	Check Gemini Pro API connection and simplify the prompt structure.
6	Dashboard Display	All summary cards show correct and consistent financial figures.	Cards show zero, null, or mismatched values.	Trace data flow from Firebase to the frontend and fix the broken data mapping.

Ways of monitoring are conducted:

a. After every new feature is merged into the main building, one team member who was not involved in building that feature performs a full walkthrough to test the website from a fresh perspective, just like in a grayscale rollout.

b. The Firebase console is checked regularly during development for any authentication errors, failed reads or writes from documents inputted by users, or storage issues.

c. Before any new code is confirmed to be merged with the main build, the developer runs through the affected feature locally to confirm it works as expected.

d. The full system is tested end-to-end by all team members at least once before the final demonstration is conducted and locked in as the Golden Release.

4.3. Assumptions:

State the key operational & environmental conditions that you assumed to be valid during the development of the system and deployment.

a. Users, as in all members, have a stable internet connection and a screen resolution on their own devices, as kaOne relies on Firebase for real-time data storage, authentication, and file uploads, as well as supporting kaOne's current responsive layout.

b. Users access kaOne through a modern web browser like Google Chrome, Microsoft Edge, or Mozilla Firefox, and ensure the browser is up to date.

c. Users possess a valid email address for the account, as kaOne's only required setup is an email address.

d. Receipt and invoice photos uploaded by users are of sufficient clarity, resolution, and have no corrupted contents to ensure that AI can accurately parse items, quantities, and costs.

e. Users are expected to have at least three months of financial data logged in the system before the Gemini Pro Forecast is expected to produce reliable results and predictions.

f. The Gemini Pro API remains accessible and responsive throughout the development period and the hackathon demonstration.

g. Users are assumed to be Malaysian individuals or micro-business owners who are familiar with Malaysian simple financial terms such as LHDN and Ringgit Malaysia (RM).

h. All members have consistent access to the shared updated codebase via version control throughout the development period, with no major conflicts that would ruin the Golden Release, as well as ensuring all members are accessing the latest development done.

4.4. External Dependencies:

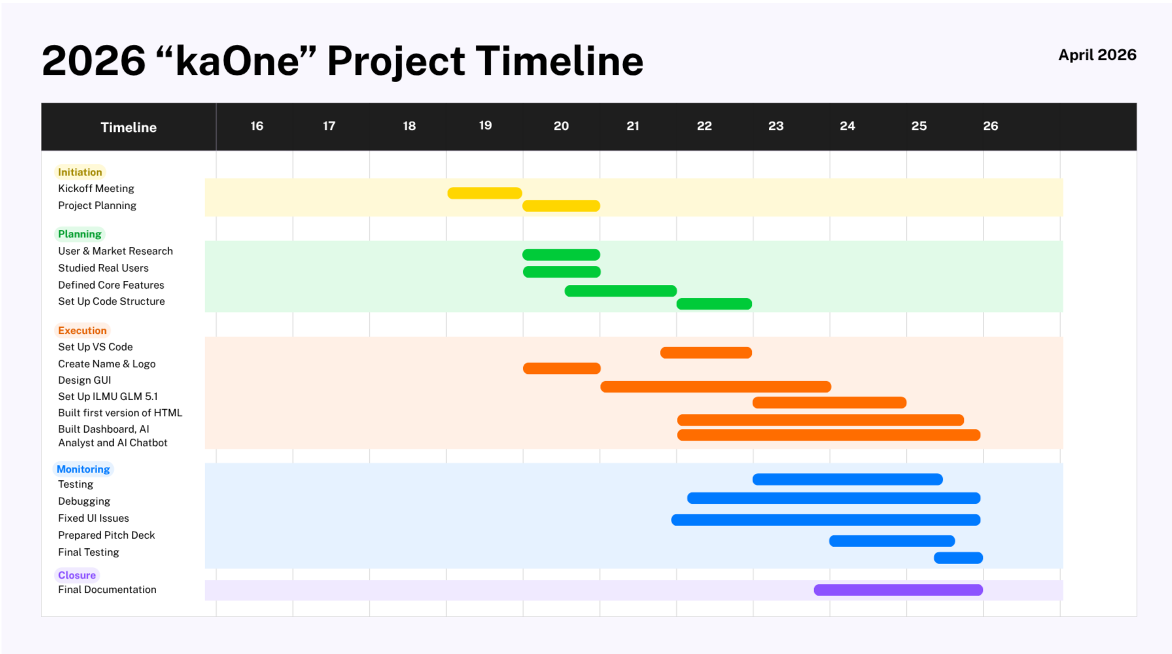
This section lists all the external tools that are applied to kaOne's system to function.

Tools	Purpose	Risks
Gemini Pro	It is the Core AI reasoning engine that powers the AI Analyst chat, processes financial state, generates personalized insights, and produces structured forecast outputs.	High – If the Gemini Pro API is inaccessible, rate-limited, or out of token availability, the AI Analyst and forecast features will fail entirely while core inventory and budget features remain accessible.
Google Firebase Firestore	Acts as an all-in-one cloud platform to handle user authentication, a Firestore database that stores all transactions, inventory, and budget data, as well as Storage for receipts and documents uploaded by users.	High - If Firebase experiences severe disruptions, the entire platform becomes inaccessible as authentication, data retrieval, and file uploads all depend on it.
Web Browser	It plays a vital role as kaOne is a web-based platform that can be run on a modern browser like Chrome, Edge, and Firefox to render the UI correctly, including dashboard cards, progress bars, forecast cards, and chat inference	Low – An outdated browser will cause rendering inconsistencies.
Internet Connection	All Firebase services and the Gemini Pro API require a stable internet connection to function and even to run the website in the first place.	Low- Unstable connection will disrupt data syncing, AI responses, and file uploads.

5. Project Management & Team Contributions

The duration of the Preliminary round is approximately 10 days, considering the two-week sprint.

5.1. Project Timeline:



5.2. Team Members Role:

Name	Role	What They Did
Nur Zahidah Husna Binti Mohd Rizal Hadi	Project Leader & Backend Development	• Led the team. • Booked mentorship sessions. • Managed backend development • Database connection • QATD documentation • Installed Flask.
Siti Nur Amalina Binti Jusoh	Frontend Development & Documentation	• Built frontend. • Generated API key. • Wrote PRD and SAD documentation. • Deployment Testing
Anjana Natalie Menon	Backend Development & Video Editing	• Finished database connection. • Recorded demo video explanation.

		• Edited pitching video. • Debugging
Husna Binti Shahar	Frontend Development & Documentation	• Built frontend. • Contributed to PRD, SAD, and pitching slides. • Backend refinement.
Batrisyia Binti Azhar	Frontend & UI Development	• Built frontend and UI. • Created pitching slides • Contributed to documentation. • Frontend refinement

Recommendations:

1. **Add API Rate Limiting and Caching**
   Every time someone asks the AI a question, it calls the ILMU model directly. If multiple users ask the same question, or the same user asks twice, the system pays for those requests each time. A simple caching layer would store responses for common questions for an hour or two. The next time someone asks the same thing, the system returns the saved answer instead of calling the model again. This cuts costs and makes responses faster.

2. **Voice Input for Analysis**
   Some people struggle with vision issues or shaky hands. Voice makes kaOne accessible to people who would otherwise give up. For elderly users running small shops. For users with dyslexia. For anyone who finds typing frustrating.

3. **Caching Dashboard Data**
   The dashboard shows things like current balance, total income, and total expenses. These numbers do not change until a new transaction is added. Caching these numbers and refreshing them only when a transaction changes would make the dashboard load faster and reduce work on the database.

4. **Real-Time Price Adjustment Alerts**
   If prices go up 20%, kaOne should know before the user buys their next shopping. Pulling from public data or supplier APIs, the AI could say "Item is RM 10 more this week than last week. Consider buying two weeks worth now or finding a different supplier." For individuals, same thing with groceries or petrol. The user saves money by buying it at the right time.

5. **Personalized Tax Rules Based on User Industry**
   We should let users select their industry during signup, then the AI learns which expense categories are tax deductible for that specific business type as they face different tax deductions. When a baker buys flour, kaOne will determine if the flour is deductible from tax.

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